

Project Development Phase

Model Performance Test

Date	02 October 2026
Team ID	SWTID-2026-8568
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Power BI model performance testing for the Auto-Ticket Classification project is recorded in the template below.

S.No.	Parameter	Screenshot / Values																																																																																																																																																									
1.	Data Rendered	Dataset: ServiceNow incident export (Tickets table) - 500 tickets, 11 columns, raised by students and teachers between 03-Aug-2026 and 30-Sep-2026. Each ticket carries only Caller Name and Short Description from the user; Category and Subcategory were assigned by the Flow Designer keyword rules. Power BI Desktop Table view Tickets (500 rows, 11 columns) <table><tr><th>Number</th><th>Caller</th><th>Short Description</th><th>Category</th><th>Subcategory</th><th>Priority</th><th>State</th><th>Opened At</th><th>Data v Tickets</th></tr><tr><td>INC0001001</td><td>Rahul Varma</td><td>WiFi not connecting in Library</td><td>Network</td><td>WiFi</td><td>3 - Moderate</td><td>Closed</td><td>03-08-2026</td><td>[] Number</td></tr><tr><td>INC0001002</td><td>Anitha Varma</td><td>Projector display flickering in Lab 2</td><td>Hardware</td><td>Projector</td><td>4 - Low</td><td>In Progress</td><td>03-08-2026</td><td>[] Caller</td></tr><tr><td>INC0001003</td><td>Harish Nair</td><td>WiFi not connecting in Auditorium</td><td>Network</td><td>WiFi</td><td>3 - Moderate</td><td>Closed</td><td>03-08-2026</td><td>[] Caller Type</td></tr><tr><td>INC0001004</td><td>Sneha Iyer</td><td>Login failed after password change</td><td>Access</td><td>Forgot Password</td><td>2 - High</td><td>Closed</td><td>03-08-2026</td><td>[] Short Description</td></tr><tr><td>INC0001005</td><td>Ishaan Menon</td><td>Projector display flickering in Lab 2</td><td>Hardware</td><td>Projector</td><td>2 - High</td><td>Closed</td><td>03-08-2026</td><td>[] Category</td></tr><tr><td>INC0001006</td><td>Divya Varma</td><td>System Hanging during online exam</td><td>Performance</td><td>Slow Computer</td><td>1 - Critical</td><td>Resolved</td><td>03-08-2026</td><td>[] Subcategory</td></tr><tr><td>INC0001007</td><td>Sneha Menon</td><td>No signal on Projector in Lab 1</td><td>Hardware</td><td>Projector</td><td>4 - Low</td><td>Resolved</td><td>03-08-2026</td><td>[] Priority</td></tr><tr><td>INC0001008</td><td>Ananya Varma</td><td>Cannot connect to school WiFi on laptop</td><td>Network</td><td>WiFi</td><td>2 - High</td><td>Closed</td><td>03-08-2026</td><td>[] State</td></tr><tr><td>INC0001009</td><td>Deepa Kumar</td><td>WiFi very weak near Room 204</td><td>Network</td><td>WiFi</td><td>3 - Moderate</td><td>Closed</td><td>03-08-2026</td><td>[] Opened At</td></tr><tr><td>INC0001010</td><td>Vikram Iyer</td><td>System Hanging during online exam</td><td>Performance</td><td>Slow Computer</td><td>3 - Moderate</td><td>Resolved</td><td>03-08-2026</td><td>[5] Resolution Hours</td></tr><tr><td>INC0001011</td><td>Anitha Varma</td><td>Projector not turning on in Room 204</td><td>Hardware</td><td>Projector</td><td>2 - High</td><td>Resolved</td><td>03-08-2026</td><td>[] Classification Source</td></tr><tr><td>INC0001012</td><td>Meera Krishnan</td><td>WiFi very weak near Room 110</td><td>Network</td><td>WiFi</td><td>3 - Moderate</td><td>Closed</td><td>03-08-2026</td><td></td></tr><tr><td>INC0001013</td><td>Priya Varma</td><td>No WiFi signal in Library</td><td>Network</td><td>WiFi</td><td>2 - High</td><td>Resolved</td><td>04-08-2026</td><td></td></tr><tr><td>INC0001014</td><td>Sneha Pillai</td><td>System Hanging during online exam</td><td>Performance</td><td>Slow Computer</td><td>4 - Low</td><td>Closed</td><td>04-08-2026</td><td></td></tr><tr><td>INC0001015</td><td>Mohan Pillai</td><td>No WiFi signal in Lab 2</td><td>Network</td><td>WiFi</td><td>3 - Moderate</td><td>Resolved</td><td>04-08-2026</td><td></td></tr><tr><td>INC0001016</td><td>Aarav Sharma</td><td>Computer very Slow in Lab 1</td><td>Performance</td><td>Slow Computer</td><td>3 - Moderate</td><td>Closed</td><td>04-08-2026</td><td></td></tr></table> Showing 16 of 500 rows Table view of the Tickets table loaded into Power BI (first 16 of 500 rows).	Number	Caller	Short Description	Category	Subcategory	Priority	State	Opened At	Data v Tickets	INC0001001	Rahul Varma	WiFi not connecting in Library	Network	WiFi	3 - Moderate	Closed	03-08-2026	[] Number	INC0001002	Anitha Varma	Projector display flickering in Lab 2	Hardware	Projector	4 - Low	In Progress	03-08-2026	[] Caller	INC0001003	Harish Nair	WiFi not connecting in Auditorium	Network	WiFi	3 - Moderate	Closed	03-08-2026	[] Caller Type	INC0001004	Sneha Iyer	Login failed after password change	Access	Forgot Password	2 - High	Closed	03-08-2026	[] Short Description	INC0001005	Ishaan Menon	Projector display flickering in Lab 2	Hardware	Projector	2 - High	Closed	03-08-2026	[] Category	INC0001006	Divya Varma	System Hanging during online exam	Performance	Slow Computer	1 - Critical	Resolved	03-08-2026	[] Subcategory	INC0001007	Sneha Menon	No signal on Projector in Lab 1	Hardware	Projector	4 - Low	Resolved	03-08-2026	[] Priority	INC0001008	Ananya Varma	Cannot connect to school WiFi on laptop	Network	WiFi	2 - High	Closed	03-08-2026	[] State	INC0001009	Deepa Kumar	WiFi very weak near Room 204	Network	WiFi	3 - Moderate	Closed	03-08-2026	[] Opened At	INC0001010	Vikram Iyer	System Hanging during online exam	Performance	Slow Computer	3 - Moderate	Resolved	03-08-2026	[5] Resolution Hours	INC0001011	Anitha Varma	Projector not turning on in Room 204	Hardware	Projector	2 - High	Resolved	03-08-2026	[] Classification Source	INC0001012	Meera Krishnan	WiFi very weak near Room 110	Network	WiFi	3 - Moderate	Closed	03-08-2026		INC0001013	Priya Varma	No WiFi signal in Library	Network	WiFi	2 - High	Resolved	04-08-2026		INC0001014	Sneha Pillai	System Hanging during online exam	Performance	Slow Computer	4 - Low	Closed	04-08-2026		INC0001015	Mohan Pillai	No WiFi signal in Lab 2	Network	WiFi	3 - Moderate	Resolved	04-08-2026		INC0001016	Aarav Sharma	Computer very Slow in Lab 1	Performance	Slow Computer	3 - Moderate	Closed	04-08-2026	
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2.	Data Preprocessing	Cleaning and shaping done in Power Query Editor: promoted headers, set data types, trimmed Short Description, removed duplicate ticket numbers, filtered out test tickets, replaced blank Category with "Other", and added Keyword Detected plus Date, Hour and Weekday columns. Column profiling shows 100% valid, 0% error, 0% empty for all key fields. Power Query Editor Tickets Data quality profile (column quality / distribution) <table><thead><tr><th>Column</th><th>Quality / Distribution</th><th>Valid</th></tr></thead><tbody><tr><td>Number</td><td>500 distinct</td><td>100% valid</td></tr><tr><td>Short Description</td><td>500 rows</td><td>100% valid</td></tr><tr><td>Category</td><td>5 distinct</td><td>100% valid</td></tr><tr><td>Subcategory</td><td>5 distinct</td><td>100% valid</td></tr><tr><td>Priority</td><td>4 distinct</td><td>100% valid</td></tr><tr><td>Opened At</td><td>DateTime</td><td>100% valid</td></tr><tr><td>Resolution Hours</td><td>Decimal</td><td>100% valid</td></tr></tbody></table> M code (Advanced Editor, excerpt) <pre>= Table.AddColumn(Trimmed, "Keyword Detected", each if Text.Contains([Short Description], "wifi", Comparer.OrdinalIgnoreCase) then "WIFI" else if Text.Contains([Short Description], "projector", Comparer.OrdinalIgnoreCase) then "Projector" else if Text.Contains([Short Description], "password") or Text.Contains([Short Description], "login") then "Password/Login" else if Text.Contains([Short Description], "slow") or Text.Contains([Short Description], "hanging") then "Slow/Hanging" else "None")</pre> APPLIED STEPS x Source (ServiceNow CSV export) x Promoted Headers x Changed Type x Trimmed Short Description x Lowercased Keyword Column x Removed Duplicates (Number) x Replaced Nulls (Category = Other) x Added Keyword Detected x Added Date / Hour / Weekday x Filtered Test Tickets x Renamed Columns	Column	Quality / Distribution	Valid	Number	500 distinct	100% valid	Short Description	500 rows	100% valid	Category	5 distinct	100% valid	Subcategory	5 distinct	100% valid	Priority	4 distinct	100% valid	Opened At	DateTime	100% valid	Resolution Hours	Decimal	100% valid																																																		
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3.	Utilization of Data Filters	Slicers used: Category, Priority, Caller Type and Opened At (date range). Example shown: Category = Network and Hardware, Priority = High and Moderate, Date = September 2026, which returns 88 of 500 tickets. A page-level filter (Category <> blank) and a visual-level Top N filter on Subcategory are also applied. Auto-Ticket Classification Filters Applied Slicers: Category = Network, Hardware Priority = 2 - High, 3 - Moderate Date = 01-Sep-2026 to 30-Sep-2026 Category ☒ Network ☒ Hardware ☐ Access ☐ Performance ☐ Other Priority ☐ 1 - Critical ☒ 2 - High ☒ 3 - Moderate ☐ 4 - Low Caller Type ☒ Student ☒ Teacher Opened At (Date slicer) 01-09-2026 - 30-09-2026 88 Filtered Tickets 4.7 Avg Resolution (hrs) 17.6% % of All Tickets Filtered tickets by Subcategory <table><thead><tr><th>Subcategory</th><th>Count</th></tr></thead><tbody><tr><td>WIFI</td><td>53</td></tr><tr><td>Projector</td><td>35</td></tr></tbody></table> Filtered tickets by Priority <table><thead><tr><th>Priority</th><th>Count</th></tr></thead><tbody><tr><td>High</td><td>44</td></tr><tr><td>Moderate</td><td>44</td></tr></tbody></table> Filtered tickets per day (September 2026) <table><thead><tr><th>Date</th><th>Count</th></tr></thead><tbody><tr><td>09-01</td><td>2</td></tr><tr><td>09-02</td><td>6</td></tr><tr><td>09-03</td><td>4</td></tr><tr><td>09-04</td><td>4</td></tr><tr><td>09-05</td><td>4</td></tr><tr><td>09-06</td><td>2</td></tr><tr><td>09-07</td><td>2</td></tr><tr><td>09-08</td><td>5</td></tr><tr><td>09-09</td><td>2</td></tr><tr><td>09-10</td><td>4</td></tr><tr><td>09-11</td><td>2</td></tr><tr><td>09-12</td><td>4</td></tr><tr><td>09-13</td><td>2</td></tr><tr><td>09-14</td><td>2</td></tr><tr><td>09-15</td><td>10</td></tr><tr><td>09-16</td><td>2</td></tr><tr><td>09-17</td><td>5</td></tr><tr><td>09-18</td><td>3</td></tr><tr><td>09-19</td><td>2</td></tr><tr><td>09-20</td><td>2</td></tr><tr><td>09-21</td><td>4</td></tr><tr><td>09-22</td><td>3</td></tr><tr><td>09-23</td><td>2</td></tr><tr><td>09-24</td><td>9</td></tr><tr><td>09-25</td><td>2</td></tr><tr><td>09-26</td><td>4</td></tr><tr><td>09-27</td><td>3</td></tr><tr><td>09-28</td><td>2</td></tr><tr><td>09-29</td><td>4</td></tr><tr><td>09-30</td><td>4</td></tr></tbody></table>	Subcategory	Count	WIFI	53	Projector	35	Priority	Count	High	44	Moderate	44	Date	Count	09-01	2	09-02	6	09-03	4	09-04	4	09-05	4	09-06	2	09-07	2	09-08	5	09-09	2	09-10	4	09-11	2	09-12	4	09-13	2	09-14	2	09-15	10	09-16	2	09-17	5	09-18	3	09-19	2	09-20	2	09-21	4	09-22	3	09-23	2	09-24	9	09-25	2	09-26	4	09-27	3	09-28	2	09-29	4	09-30	4
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4.	DAX Queries Used	Measures and calculated column created in the Tickets model: Total Tickets = COUNTROWS (Tickets) Auto Classified Tickets = CALCULATE ([Total Tickets], Tickets[Classification Source] = "Flow Designer") Auto Classification Rate % = DIVIDE ([Auto Classified Tickets], [Total Tickets]) Avg Resolution Hours = AVERAGE (Tickets[Resolution Hours]) Open Tickets =																																																																										

S.No.	Parameter	Screenshot / Values
		CALCULATE ([Total Tickets], Tickets[State] IN { "New", "In Progress" }) Category Share % = DIVIDE ([Total Tickets], CALCULATE ([Total Tickets], ALL (Tickets[Category]))) Tickets MTD = TOTALMTD ([Total Tickets], 'Date'[Date]) Manual Effort Saved (hrs) = [Auto Classified Tickets] * 3 / 60 Keyword Detected (calculated column) = SWITCH (TRUE (), SEARCH ("wifi", Tickets[Short Description], 1, 0) > 0, "WiFi", SEARCH ("projector", Tickets[Short Description], 1, 0) > 0, "Projector", SEARCH ("password", Tickets[Short Description], 1, 0) > 0 SEARCH ("login", Tickets[Short Description], 1, 0) > 0, "Password/Login", SEARCH ("slow", Tickets[Short Description], 1, 0) > 0 SEARCH ("hanging", Tickets[Short Description], 1, 0) > 0, "Slow/Hanging", "None")
5.	Dashboard design	No of Visualizations / Graphs - 10 5 KPI cards (Total Tickets, Auto-Classified, Auto-Classification Rate, Avg Resolution Hours, Open Tickets), donut chart (tickets by Category), bar chart (tickets by Subcategory), column chart (tickets by Priority), line chart (weekly trend by Category) and column chart (ticket status). In total the dashboard page holds 10 visuals (5 cards + 5 charts).

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6.	Report Design	No of Visualizations / Graphs - 4 Report page 2: stacked column chart (monthly tickets by Category), bar chart (average resolution time by Subcategory), heat map matrix (weekday x Category) and a keyword-to-category table that shows how many tickets each Flow Designer rule classified. IT Helpdesk Detailed Report - Classification & Resolution Analysis Page 2 of 2 Monthly tickets by Category (stacked) <table><tr><th>Month</th><th>Network</th><th>Hardware</th><th>Access</th><th>Performance</th><th>Other</th></tr><tr><td>Aug 2026</td><td>92</td><td>53</td><td>56</td><td>41</td><td>9</td></tr><tr><td>Sep 2026</td><td>88</td><td>47</td><td>69</td><td>40</td><td>11</td></tr></table> Avg resolution time (hrs) by Subcategory <table><tr><th>Subcategory</th><th>Avg resolution time (hrs)</th></tr><tr><td>Projector</td><td>6.5</td></tr><tr><td>Slow Computer</td><td>4.9</td></tr><tr><td>WiFi</td><td>3.1</td></tr><tr><td>Forgot Password</td><td>1.2</td></tr></table> Tickets by Weekday x Category (heat map) <table><tr><th></th><th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th><th>Sat</th></tr><tr><td>Access</td><td>25</td><td>29</td><td>19</td><td>11</td><td>17</td><td>24</td></tr><tr><td>Hardware</td><td>19</td><td>18</td><td>22</td><td>14</td><td>16</td><td>11</td></tr><tr><td>Network</td><td>36</td><td>20</td><td>39</td><td>27</td><td>39</td><td>19</td></tr><tr><td>Performance</td><td>22</td><td>19</td><td>9</td><td>7</td><td>11</td><td>13</td></tr></table> Top keyword triggers (Flow Designer rules) <table><tr><th>Keyword</th><th>Mapped to</th><th>Tickets</th></tr><tr><td>WiFi</td><td>Network / WiFi</td><td>180</td></tr><tr><td>Projector</td><td>Hardware / Projector</td><td>100</td></tr><tr><td>Password / Login</td><td>Access / Forgot Password</td><td>125</td></tr><tr><td>Slow / Hanging</td><td>Performance / Slow Computer</td><td>81</td></tr><tr><td>(no keyword)</td><td>Other / Manual Review</td><td>14</td></tr></table>	Month	Network	Hardware	Access	Performance	Other	Aug 2026	92	53	56	41	9	Sep 2026	88	47	69	40	11	Subcategory	Avg resolution time (hrs)	Projector	6.5	Slow Computer	4.9	WiFi	3.1	Forgot Password	1.2		Mon	Tue	Wed	Thu	Fri	Sat	Access	25	29	19	11	17	24	Hardware	19	18	22	14	16	11	Network	36	20	39	27	39	19	Performance	22	19	9	7	11	13	Keyword	Mapped to	Tickets	WiFi	Network / WiFi	180	Projector	Hardware / Projector	100	Password / Login	Access / Forgot Password	125	Slow / Hanging	Performance / Slow Computer	81	(no keyword)	Other / Manual Review	14
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